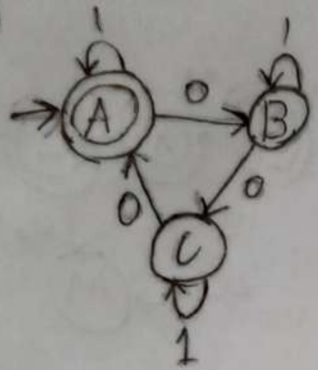
Masrur Sarar

ID: 23201003

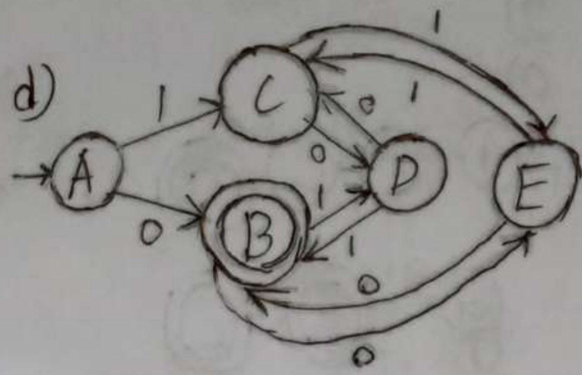
Assignment → 1 CSE 331

1)

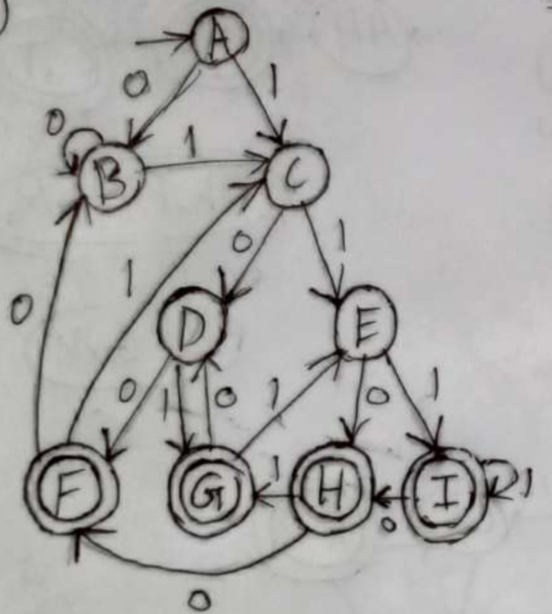
a)



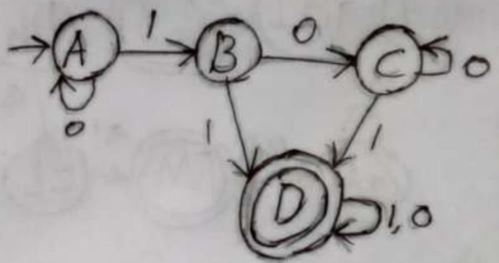
d)



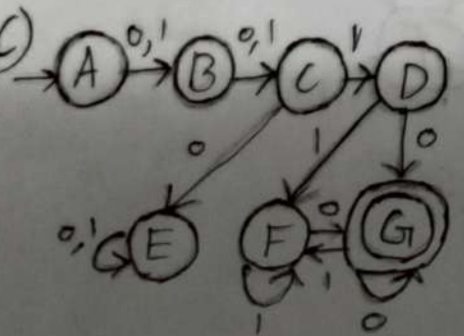
b)



e)

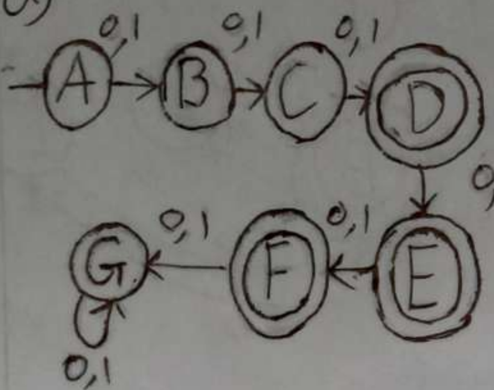


c)

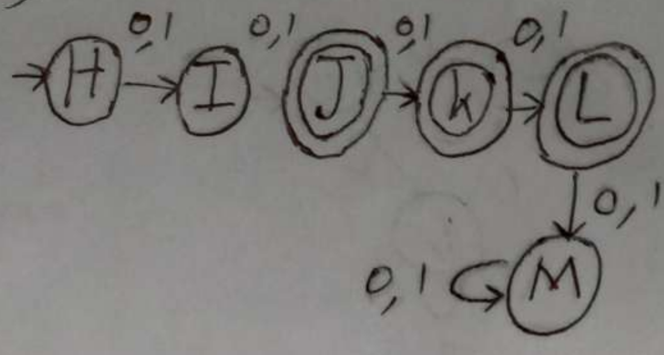


2)

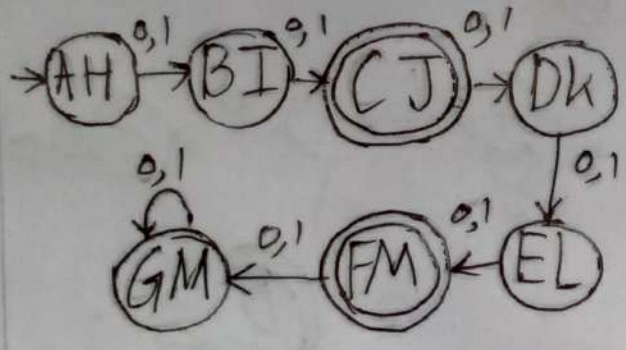
a)



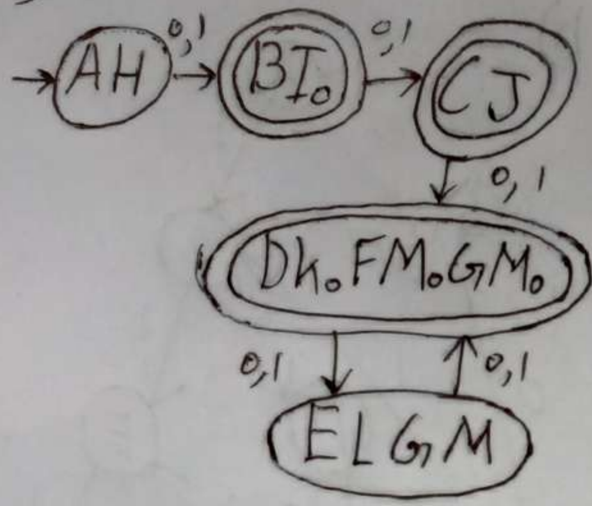
b)



c) $A \Delta B$



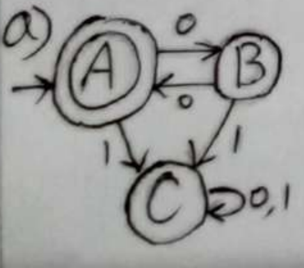
e)



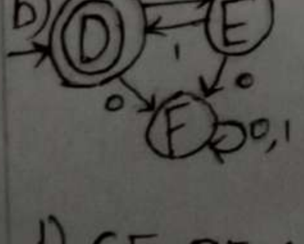
d) $7 \times 6 \times 2 = 84$ states

3)

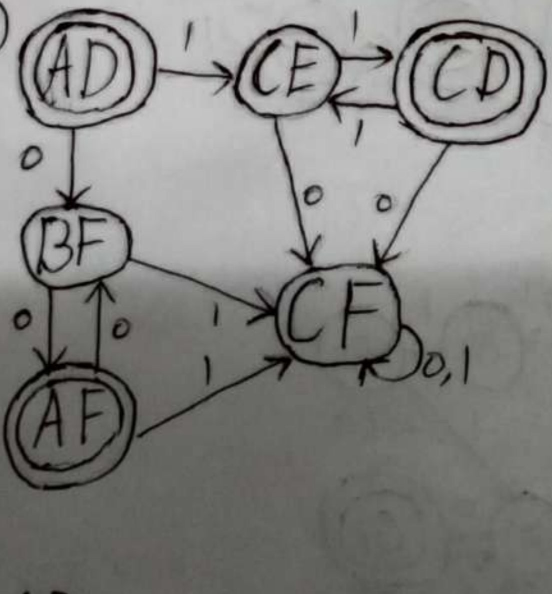
a)



b)



c)



d) CE, BF, CF, AF, CD

e) AF, CD

4) a) For NFA

Total stage, $n=5$

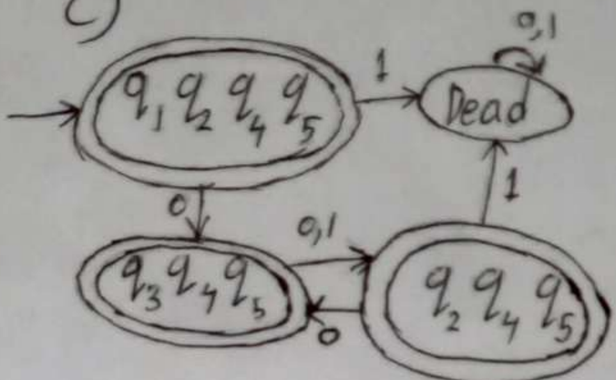
rejecting stage, $r=3$

For DFA,

Max rejected stage,

$= 2^{3} = 8$

c)

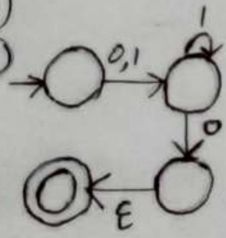


b) ε-closure of q1 is

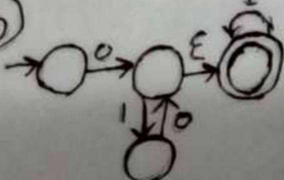
{q1, q2, q4, q5}

5)

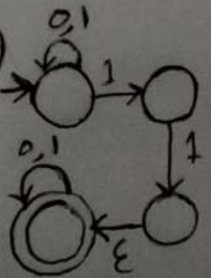
a)



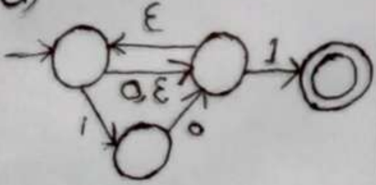
b)



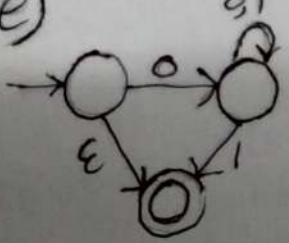
c)



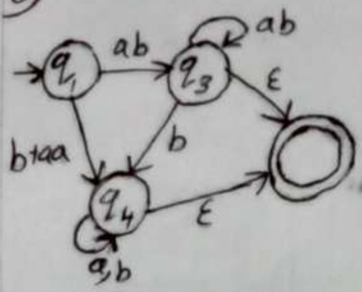
d)

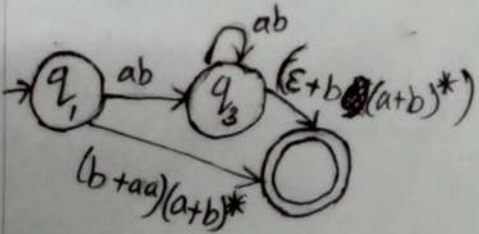


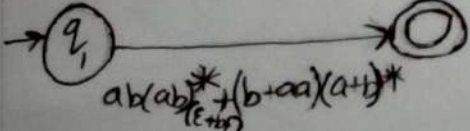
e)



6)







~~Ans:- ab(ab)*+(b+aa)(a+b)*~~

~~Ans:- ab(ab)*(ε+bε)+(b+aa)(a+b)*~~

Ans:- ab(ab)*(ε+b(a+b)*)+(b+aa)(a+b)*